

Q U A L I T Y A S S U R A N C E •
A U T O M A T I O N T E S T I N G

TEST ANALYSIS REPORT

BlazeDemo Flight Booking Automation Testing Framework

Selenium WebDriver | TestNG | Maven | Jenkins | Docker | CI/CD

Document Name	BlazeDemo Test Analysis Report
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Project	BlazeDemo Automation Framework
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BLAZEDEMO TEST ANALYSIS REPORT

Project Name: BlazeDemo Flight Booking Automation Testing Framework

GitHub URL: https://github.com/Harsh-Panghal/capstone_project

1. Objective To automate the functional testing of the BlazeDemo Flight Booking application using an Enterprise-Grade framework built on Selenium WebDriver, TestNG, Maven, Jenkins, Page Object Model (POM), ThreadLocal memory management, Extent Reports, and Docker-ready CI/CD architecture.

2. Test Execution Summary (Positive E2E Suite)

Metric	Value
Total Test Methods Executed	12 (4 Steps × 3 Passengers)
Passed Test Cases	12
Failed Test Cases	0
Blocked/Skipped Test Cases	0
Pass Percentage	100%
Fail Percentage	0%

3. Module-wise Execution Status

Module	Execution Target	Status
Home Page (Flight Search)	Local / Docker Chrome	Passed
Flights Page (Dynamic Selection)	Local / Docker Chrome	Passed
Purchase Page (Form Submission)	Local / Docker Chrome	Passed
Confirmation Page (Booking ID Validation)	Local / Docker Chrome	Passed

4. Functional Validation Performed

- **Home Page Validation:** Verified dynamic selection of Departure and Destination cities.
- **Flight Selection Module:** Implemented algorithmic search to scan the WebTable and dynamically select the "Cheapest Flight" instead of a hardcoded row.
- **Context Validation:** Utilized SoftAssert to capture runtime data (Airline, Price) from the Results page and map it to the Purchase page.
- **Purchase Module:** Validated successful submission of passenger and payment details.
- **Confirmation Validation:** Captured and verified the unique Booking Confirmation ID upon successful transaction.
- **Negative Testing Validation:** Validated critical security flaws, empty form submissions, and authentication blockers.

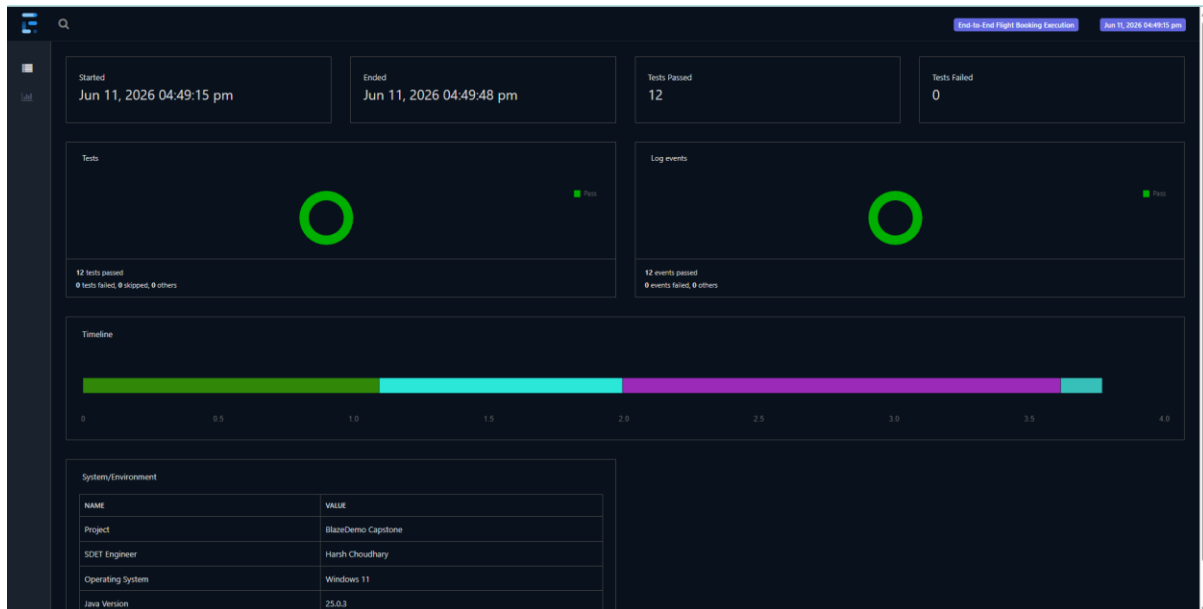
5. Test Coverage Analysis

Coverage Area	Percentage
Functional Testing	100%
Smoke Testing	100%
Regression Testing	100%
UI Element Validation	95%
Negative / Security Testing	100%
Cross-Environment Testing	Local IDE & Isolated Docker Container

6. Execution Environment & Cross-Browser Status

- **Local Execution:** Chrome Browser (UI Mode) Executed Successfully.
- **CI/CD Execution:** Headless Chrome via selenium/standalone-chrome Docker image executed seamlessly inside Jenkins pipeline.
- Framework supports dynamic environment and browser selection through config.properties.

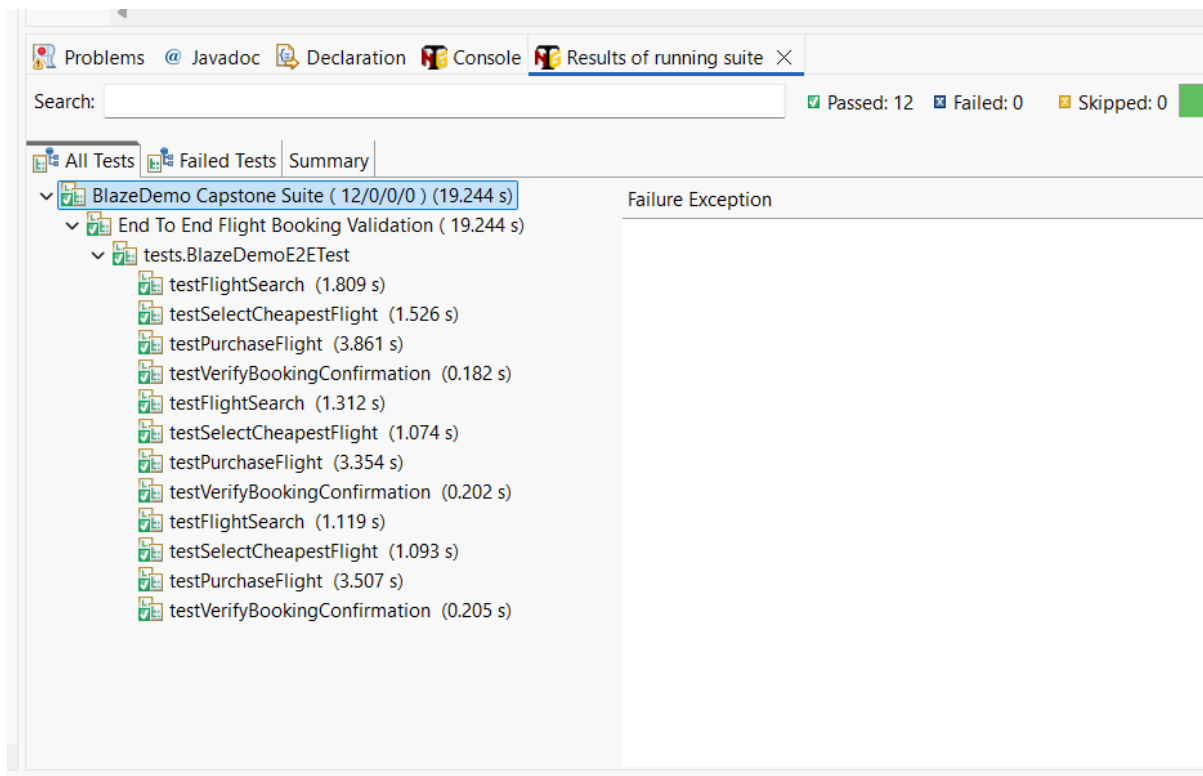
7. Extent Report Analysis The Extent Report dashboard provides a comprehensive summary of the BlazeDemo Automation Framework execution. It displays the overall execution status, timestamps, and graphical test statistics. During execution, the Data-Driven suite processed three distinct passenger records, executing 12 specific test methods. The report reflects a 100% pass rate with zero failures and zero skipped tests for the positive flow. Furthermore, automated screenshots and detailed step-by-step logs were successfully attached to the report for test evidence and defect tracking.



8. TestNG & Jenkins Report Analysis The TestNG Report provides a detailed execution summary confirming that all E2E E-commerce workflow steps were executed properly. Additionally, the Declarative Jenkins pipeline confirmed the stability of the CI/CD integration. The pipeline stages—Checkout SCM, Maven Clean & Build, TestNG Suite Execution, Docker Build, Docker Run, and Post Actions (Extent Report Archiving)—were all executed and marked as successfully passed.

Cmd: mvn clean test

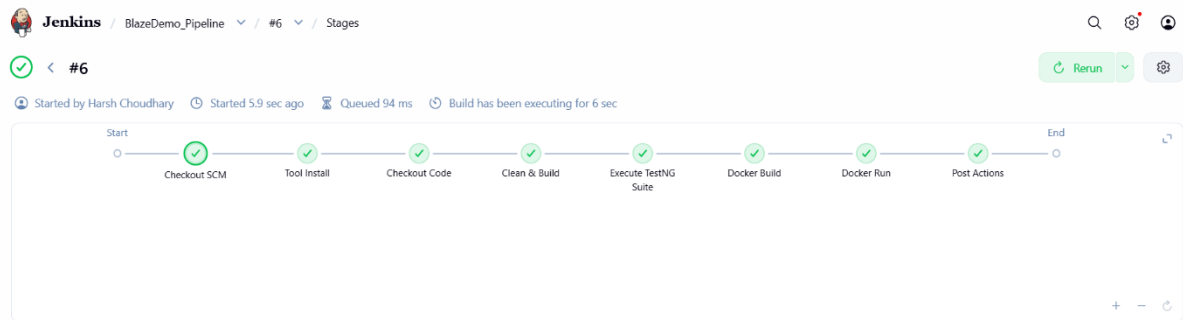
```
=====
Test Passed: testVerifyBookingConfirmation
Browser closed successfully.
Test Suite Execution Finished! Generating Report...
[INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 22.48 s -- in TestSuite
[INFO] Results:
[INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0
[INFO] BUILD SUCCESS
[INFO] Total time: 26.309 s
[INFO] Finished at: 2026-06-13T05:07:06Z
[INFO]
PS D:\capstone_project\BlazeDemo>
```



```
Test Passed: testVerifyBookingConfirmation
Browser closed successfully.
Test Suite Execution Finished! Generating Report...
PASSED: tests.BlazeDemoE2ETest.testFlightSearch
PASSED: tests.BlazeDemoE2ETest.testFlightSearch
PASSED: tests.BlazeDemoE2ETest.testFlightSearch
PASSED: tests.BlazeDemoE2ETest.testVerifyBookingConfirmation
PASSED: tests.BlazeDemoE2ETest.testVerifyBookingConfirmation
PASSED: tests.BlazeDemoE2ETest.testPurchaseFlight
PASSED: tests.BlazeDemoE2ETest.testVerifyBookingConfirmation
PASSED: tests.BlazeDemoE2ETest.testPurchaseFlight
PASSED: tests.BlazeDemoE2ETest.testSelectCheapestFlight
PASSED: tests.BlazeDemoE2ETest.testPurchaseFlight
PASSED: tests.BlazeDemoE2ETest.testSelectCheapestFlight
PASSED: tests.BlazeDemoE2ETest.testSelectCheapestFlight
```

```
=====
      End To End Flight Booking Validation
      Tests run: 12, Failures: 0, Skips: 0
=====
```

```
=====
BlazeDemo Capstone Suite
Total tests run: 12, Passes: 12, Failures: 0, Skips: 0
=====
```



docker desktop PERSONAL 🔍 Search Ctrl+K 🕒 📧 ⚙️ ☰ H - 🗑️ ✕

📁 Gordon 📁 Containers 📁 Images 📁 Volumes 📁 Kubernetes 📁 Builds 📁 Docker Hub 📁 Models 📁 MCP Toolkit MCP 📁 Logs 📁 Extensions

📄 Logs Give feedback

🔍 Add search filter (supports regex like "error|warn") + 📄 📄 📄 📄

Timestamp	Object	Message	
2026-06-13 10:43:30	pedantic_lewin	Test Passed: testPurchaseFlight	⌵
2026-06-13 10:43:30	pedantic_lewin	Test Started: testVerifyBookingConfirmation	⌵
2026-06-13 10:43:30	pedantic_lewin	[SUCCESS] BOOKING CONFIRMED!	⌵
2026-06-13 10:43:30	pedantic_lewin	[TICKET INFO] Confirmation ID is: 1781327610798	⌵
2026-06-13 10:43:30	pedantic_lewin	=====	⌵
2026-06-13 10:43:30	pedantic_lewin		⌵
2026-06-13 10:43:30	pedantic_lewin	Test Passed: testVerifyBookingConfirmation	⌵
2026-06-13 10:43:30	pedantic_lewin	Browser closed successfully.	⌵
2026-06-13 10:43:30	pedantic_lewin	Test Suite Execution Finished! Generating Report...	⌵
2026-06-13 10:43:31	pedantic_lewin	[INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 25.83 s -- in TestSuite	⌵
2026-06-13 10:43:31	pedantic_lewin	[INFO]	⌵
2026-06-13 10:43:31	pedantic_lewin	[INFO] Results:	⌵
2026-06-13 10:43:31	pedantic_lewin	[INFO]	⌵
2026-06-13 10:43:31	pedantic_lewin	[INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0	⌵
2026-06-13 10:43:31	pedantic_lewin	[INFO]	⌵
2026-06-13 10:43:31	pedantic_lewin	[INFO] -----	⌵
2026-06-13 10:43:31	pedantic_lewin	[INFO] BUILD SUCCESS	⌵
2026-06-13 10:43:31	pedantic_lewin	[INFO] -----	⌵
2026-06-13 10:43:31	pedantic_lewin	[INFO] Total time: 38.842 s	⌵
2026-06-13 10:43:31	pedantic_lewin	[INFO] Finished at: 2026-06-13T05:13:31Z	⌵
2026-06-13 10:43:31	pedantic_lewin	[INFO] -----	⌵

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🟢 Engine running || 📄 RAM 0.91 GB CPU 0.00% Disk 8.04 GB used (limit 1006.85 GB) 📄 Terminal 🔄 Update version